

H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

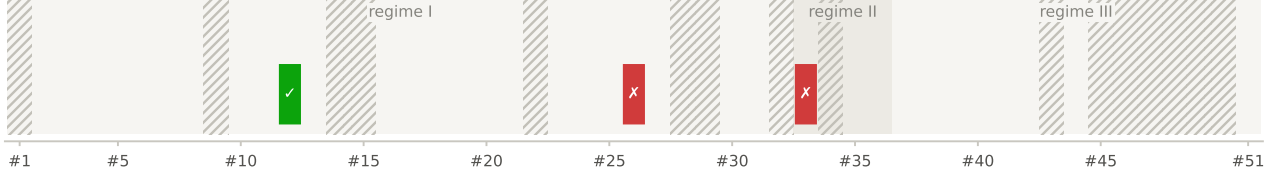
Complete 50%

Faithfulness 2/5

Usefulness 2/5

Bottom line. In what they talk about the debaters form a ferromagnet in the motion's topic; in how they reply, the drafted teams form two sublattices that the assignment switches on and the verdict switches off, and no camps form where nobody assigns sides.

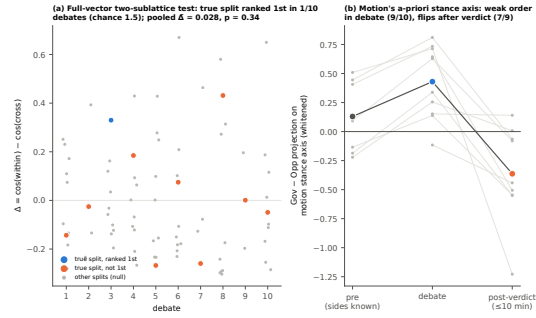
level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 2 failed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

Tested on #12: 10 debates with their own judge, motion and re-drafted teams (labels equal DQ6's ground truth). **In content the pre-registered two-sublattice test failed:** within- minus cross-team cosine $\bar{\Delta} = 0.028$ ($p = 0.34$), 1/10 teams recovered; **round 1b** repeats this with a second embedding model, style-residualized and deduplicated inputs ($\bar{\Delta} - 0.05$ to $+0.03$). Only a faint tilt along the motion's support-minus-oppose axis remains (bge $p = 0.002$, gte $p = 0.07$). **In reply stance (DQ2) the teams do order:** teammates $+0.41$, opponents -0.32 ; $\Delta = 0.72$ in 10/10 debates, beyond a calibrated agent-field null ($p = 0.002$); teams recovered in 8/10; the same pair turns hostile only when drafted apart (16/18); the order is gone after the verdict. Where nobody assigns sides (#26 vote, #33 news debate) no camps form. #34 holdout not run.



Observables. (a) Per debate, within-minus-cross cosine for every possible team split (gray) and the true split (large dot; blue if best): the true split is unremarkable, so no two-sublattice order in full content vectors. (b) On the motion's a-priori stance axis Government sits slightly on the support side during debates (9/10) and, post hoc, swings to the other side after the verdict (7/9).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Practical lesson: look for factions in reply stance, not in embeddings. Clustering messages finds labs and topics, never even the assigned teams; a cheap zero-shot stance label on reply pairs, read in aggregate against an agent-field null, recovers 8 of 10 drafted teams. The flip side matters as much: agents built on these models do not form camps on their own, not over a contested vote and not in a debate about the news. Opposition appears where a protocol assigns it and disappears when the protocol ends. For an operator, that makes a stance-based faction alarm useful mainly as a check that assigned roles are being played, and a sign of something unusual when it fires unassigned.

PHYSICAL MODEL

Models 11, 10, 01. Spin $\mathbf{s}_i$: debater i 's masked, whitened ($n = 32$), agent-centred chat mean in a debate. Sublattices $A = \text{Government}$ ($\varepsilon_i = +1$), $B = \text{Opposition}$. Fields: motion topic $h_u \hat{g}$ (uniform), assigned sides $h_s \hat{a}$ (staggered), family $\mathbf{h}_i$, verdict h_v toward winner W . Couplings k_{in} (team ferro), k_{out} (cross antiferro). ρ : correlation of sublattice fluctuations.

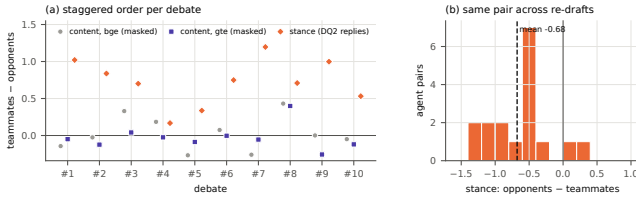
$$H = -\frac{k_{\text{in}}}{N} \sum_{i \sim j} \mathbf{s}_i \cdot \mathbf{s}_j + \frac{k_{\text{out}}}{N} \sum_{A \times B} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i (h_u \hat{g} + h_s \varepsilon_i \hat{a} + h_v \varepsilon_W \hat{a} + \mathbf{h}_i) \cdot \mathbf{s}_i$$

$$\bar{\Delta} = \langle \cos \rangle_{\text{within}} - \langle \cos \rangle_{\text{cross}}, \quad \rho = -k_{\text{out}} / (1 - k_{\text{in}})$$

H21-debate-antiferromagnet · generated 2026-10-04 10:55 by in-fra/summaries/build_summaries.py · card hypotheses/H21-debate-antiferromagnet/README.md · ratings: RE-C2 round-1b agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 content order	$\bar{\Delta}$ 0.028 → $-0.05..+0.03$ (2 models, 5 × inputs)	
P2 teams from content	1/10 → 0-1/10	×
P4d text-axis tilt	σ 0.092 (p 0.002) → gte 0.055 (p 0.07)	~
P7 AF fluctuations	ρ +0.53 → $+0.02..+0.53$ (never negative)	×
S1 stance order	Δ 0.72, 10/10; agent-field p 0.002	✓
S2 teams from stance	8/10 (chance 1.0)	✓
S3 off after verdict	-0.01 vs 0.67 during debates	✓
G12 within-pair stance (native)	-0.68 (16/18); content $\approx$ 0	✓
G26 / G33 (na- no ballot camps; no unassigned camps)	×	(as predicted)



Second figure. (a) Teammates minus opponents per debate: content cosine in two embedding models (gray, violet) scatters around zero; reply stance (orange) is positive in every debate. (b) The same agent pair across re-drafts: stance as opponents minus teammates is negative for 16 of 18 pairs.

GOAL PERIODS IN DETAIL

G12 ✓ 1b: supported in stance (two-sublattice order; content...
 G26 × failed
 G33 × failed

FAITHFULNESS SCORECARD

A mapping 1 F identifiability 2
 B assumptions 0 G ground truth 2
 C adequacy 2 H comparative 1
 D unfitted predictions 1 I transfer 0
 E interventional 1

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Stance labels are zero-shot (DQ2: pair-level “opposes” precision 0.08), so only aggregates are read, and the agent-field null assumes replies independent given agents. Stance measures agreement of positions, so an assigned staggered field alone predicts the order: coupling beyond the assignment is not shown (content fluctuations stay positive). Seven agents, four days, regime I. #26 has 9 voters with near-identical ballots (low power). #34 untouched.

ROUND 2

Test whether stance order needs the protocol: debates where sides are self-chosen, or a role switch mid-debate; a reply-level kinetic model to separate a field from coupling.

- **Essence.** LLM agents don’t form coalitions; assigned roles leave no lasting imprint.
- **H21-R1** Role imprint lifetime: roles bias stance only while active, then reverse (HH127); measure across all role periods.
- **H21-R2** Agreement is the default stance (sycophancy as ferromagnetic stance coupling); disagreement appears only when assigned.